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22nd
Asian Breast Disease Association
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Annual Scientific Meeting
Strategising Multidisciplinary
Management of Breast Diseases

IMAGING CHARACTERISTICS OF BI-RADS 6 TRIPLE-NEGATIVE BREAST CANCER WITH METASTATIC DISEASE IN A YOUNG WOMAN: A CASE OF RAPID PROGRESSION



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BACKGROUND

Triple-negative breast cancer (TNBC) is an aggressive malignancy that can mimic benign disease on ultrasound, particularly when central necrosis produces a complex cystic-solid appearance or coexisting simple cysts create a misleading benign impression, potentially delaying diagnosis and worsening outcomes.

OBJECTIVES

To describe a young woman whose grade-3 TNBC mimicked benign disease on ultrasound yet spread rapidly to nodes and pleura, and to highlight the imaging, BI-RADS, and cardiac-monitoring lessons.

CASE PRESENTATION

- 24-year-old woman with an enlarging left-breast lump. No hormonal contraception, no family history, normal cardiovascular exam
- Ultrasound (left breast): **solid mass** with an internal cystic component, up to 2.81 cm classified as BI-RADS 6 by biopsy-proven, **several simple cysts** (≤ 0.65 cm) classifies BI-RADS 2, and **enlarged left axillary nodes** up to 2.68 cm.
- Core-needle biopsy confirmed invasive carcinoma malignant and referred for staging also start anthracycline-based CEF chemotherapy regime.

IMAGING & PATHOLOGY FINDINGS

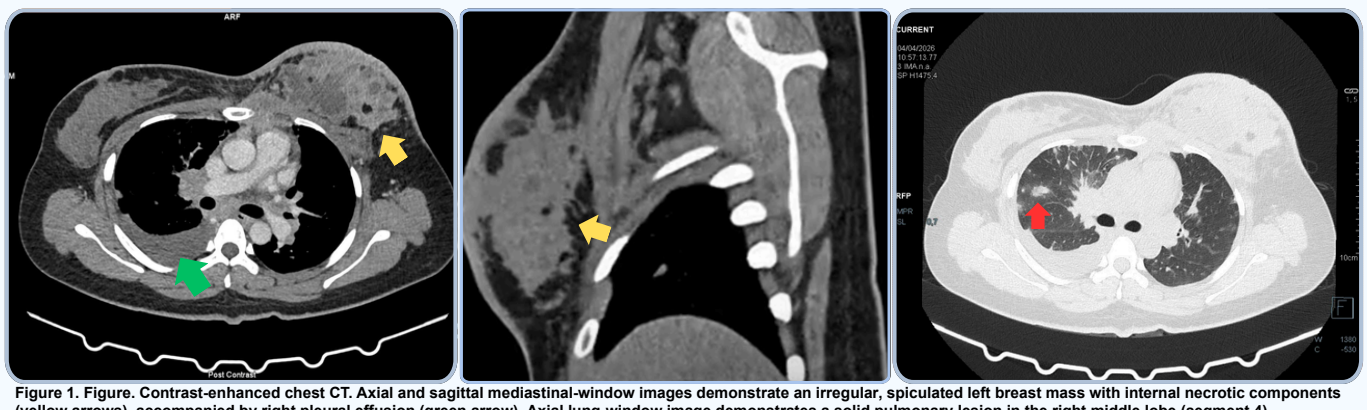


Figure 1. Figure. Contrast-enhanced chest CT. Axial and sagittal mediastinal-window images demonstrate an irregular, spiculated left breast mass with internal necrotic components (yellow arrows), accompanied by right pleural effusion (green arrow). Axial lung-window image demonstrates a solid pulmonary lesion in the right middle lobe (segment 4), suspicious for pulmonary metastasis (red arrow).

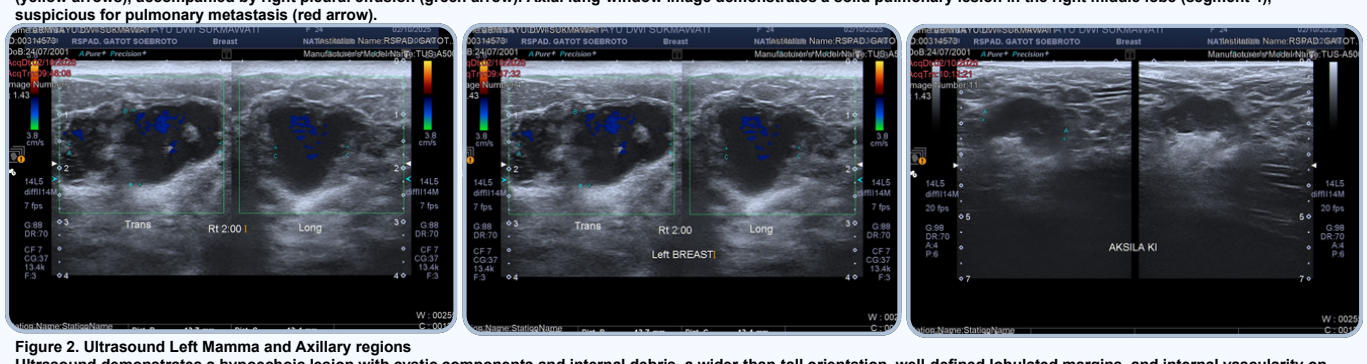


Figure 2. Ultrasound Left Mamma and Axillary regions
Ultrasound demonstrates a hypoechoic lesion with cystic components and internal debris, a wider-than-tall orientation, well-defined lobulated margins, and internal vascularity on Color Doppler. Conglomerated left axillary lymphadenopathy with loss of the fatty hilum is also noted, suspicious for metastatic nodal involvement.

Histopathology

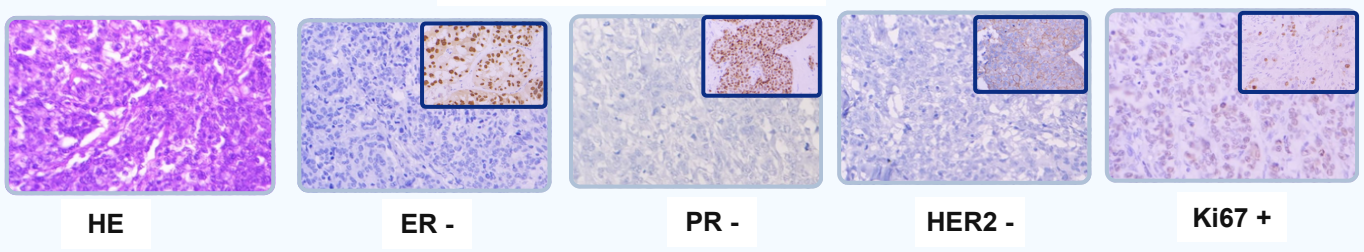


Figure 3. Immunohistochemistry (IHC) demonstrates absent estrogen receptor (ER), progesterone receptor (PR), and HER2 expression, with appropriate positive controls, consistent with a triple-negative breast cancer phenotype.

Key findings

- **Imaging:** Complex cystic and solid mass measuring ≤2.81 cm classified as BI-RADS 6, separate simple cyst measuring ≤0.65 cm classified as BI-RADS 2, and left axillary lymph node measuring ≤2.68 cm indicating node-positive disease.
- **Pathology:** Invasive carcinoma of no special type, Nottingham grade 3, with approximately 18 abnormal mitoses per 10 HPF.
- **Receptors and stage:** ER negative, PR negative, HER2 negative, and Ki-67 20 to 30%, consistent with triple-negative breast cancer, staged as cT3Nx.
- **Treatment course:** During the third cycle of CEF, left pleural effusion developed and LVEF decreased to 58%.

CONCLUSION

1. Main finding

Grade 3 TNBC mimicked benign disease on ultrasound as a complex cystic and solid mass among simple cysts, yet demonstrated rapid progression with nodal and pleural involvement within weeks.

2. Clinical meaning

Any solid or complex cystic component warrants tissue diagnosis despite reassuring imaging features, while BI-RADS 6 should be reserved for biopsy-proven malignancy and coexisting benign cysts should be accurately characterized.

3. Next step

Contrast-enhanced breast MRI should be used to better characterize complex cystic-solid lesions, complemented by FDG-PET/CT for early detection of treatment non-response during neoadjuvant chemotherapy in aggressive TNBC, alongside tissue confirmation of suspected metastatic foci.

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ACKNOWLEDGEMENTS

* The authors would like to acknowledge the Department of Radiology and the multidisciplinary team involved in the diagnosis and management of this case._
* No specific funding was received for this study.
* The authors declare no conflicts of interest.